Project: Worked examples
Title: Voltage drop and I²R loss
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Voltage drop along a circuit in volts and per cent of the nominal voltage, the conductor power loss, and an estimate of the conductor operating temperature, for DC, single-phase and three-phase AC, from resistivity and cross-section (with temperature correction), from tabulated R and X per kilometre, or from a mV/A/m figure; parallel conductors per phase; the drop limit is a user input (the regulatory limits live in BS 7671 Appendix 4 and Regulation 525 and are not embedded).

WHAT IT CALCULATES	RELATION
Conductor resistance at its operating temperature, for n conductors in parallel	$R = \rho_T L/(A\,n)$, with $\rho_T = \rho_{20}\,[1 + \alpha\,(T_c - 20)]$
Resistance and reactance from per-length figures instead	$R = R' L/n$; $X = X' L/n$
Voltage drop on a DC circuit	$\Delta V = 2\,I\,R$
Voltage drop on a single-phase AC circuit	$\Delta V = 2\,I\,(R\cos\varphi + X\sin\varphi)$
Voltage drop on a three-phase AC circuit	$\Delta V = \sqrt{3}\,I\,(R\cos\varphi + X\sin\varphi)$
Voltage drop from a tabulated mV/A/m figure, which already carries the 2 or √3	$\Delta V = (\text{mV/A/m})\,I\,L / 1000$
Power lost in the cable	$P_{loss} = 2\,I^2\,R$ (DC and single-phase); $3\,I^2\,R$ (three-phase)
Estimated conductor temperature at this current	$T_{c,est} = T_{amb} + (T_{rated} - T_{amb})(I/I_z)^2$

Basis: Voltage drop in consumers' installations[1](525); Voltage drop (mV/A/m tables and the temperature-correction method)[1](Appendix 4, section 6); Resistance, resistivity and temperature coefficient[2](Ch. on DC and AC circuits).

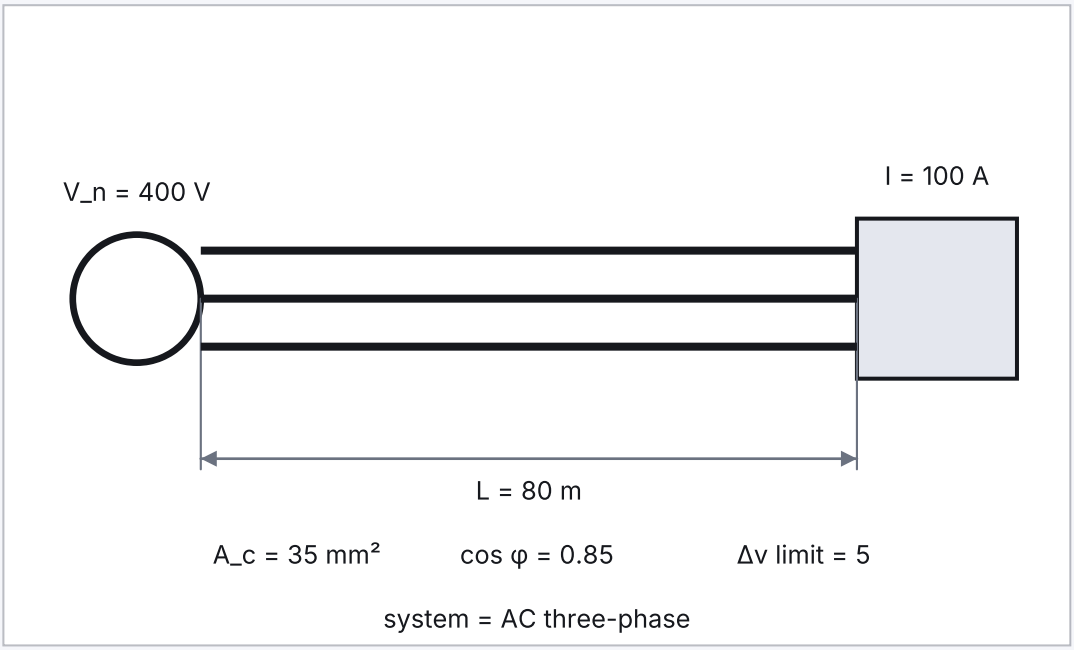


Figure 1 — A three-phase circuit: the supply, three line conductors of route length L, and the load current I. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
system	System	AC three-phase		assumption: Design convention.	any
V_n	Nominal voltage (line-to-line for three-phase)	400	V	assumption: Design convention.	12 … 1000 V
I	Design current	100	A	spec: Motor schedule MS-01: 100 A full-load current	0.1 … 5000 A
L	Route length (one way)	80	m	drawing: DRG-E-120 rev B	0.1 … 5000 m
cos_phi	Power factor (AC)	0.85		assumption: Design convention.	0.5 … 1
n_par	Parallel conductors per phase	1		assumption: Design convention.	1 … 6
data	Conductor data from	resistivity and area		assumption: Design convention.	any
rho_20	Resistivity at 20 °C (copper 0.01724, aluminium 0.0282 Ω·mm²/m)	0.01724	ohm·mm²/m	reference: Reference record: annealed copper 0.01724 Ω·mm²/m, 0.00393/K (Hughes)	0.01 … 0.05 ohm·mm²/m
alpha	Temperature coefficient of resistance	0.00393		reference: Reference record: annealed copper 0.01724 Ω·mm²/m, 0.00393/K (Hughes)	0.003 … 0.0045

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
A_c	Conductor cross-section	35	mm²	spec: Cable schedule CS-01: 35 mm² Cu XLPE/SWA	1 … 1000 mm²
T_c	Conductor operating temperature for the resistance	70	°C	assumption: Design convention.	20 … 90 °C
X_km	Reactance per km per conductor	0.08	ohm/km	spec: Cable datasheet: 0.08 Ω/km	0 … 1 ohm/km
dv_limit	Voltage-drop limit	5		spec: Project spec ES-3 §4: 5 % (BS 7671 Appendix 4 §6.4 default for other uses)	1 … 10
I_z	Tabulated current-carrying capacity for the installed conditions (temperature estimate)	119	A	spec: Cable schedule CS-01: I_z = 119 A for the installation method	1 … 5000 A
T_rated	Conductor rated temperature	70	°C	assumption: Design convention.	60 … 90 °C
T_amb	Ambient temperature	30	°C	assumption: Design convention.	0 … 60 °C

3. Assumptions

A1. Balanced load; conductor reactance from the stated X′ (zero when using resistivity alone unless X_km is given); the drop is the arithmetic (R cos φ + X sin φ) form. (Standard engineering approximation.)
A2. A mV/A/m figure is used as tabulated, with its own temperature and system convention. (Cable table convention.)
A3. The conductor temperature estimate scales the rated rise with (I/I_z)²; I_z is the user's tabulated value for the installation method and corrections. (BS 7671 Appendix 4 approach, formula only.)
A4. system (System) is taken as AC three-phase. (Design convention.)
A5. V_n (Nominal voltage (line-to-line for three-phase)) is taken as 400 V. (Design convention.)
A6. cos_phi (Power factor (AC)) is taken as 0.85. (Design convention.)
A7. n_par (Parallel conductors per phase) is taken as 1. (Design convention.)
A8. data (Conductor data from) is taken as resistivity and area. (Design convention.)
A9. T_c (Conductor operating temperature for the resistance) is taken as 70 degC. (Design convention.)
A10. T_rated (Conductor rated temperature) is taken as 70 degC. (Design convention.)
A11. T_amb (Ambient temperature) is taken as 30 degC. (Design convention.)

4. Working

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[9] sin_phi = sqrt(1 - cos_phi^2)
= sqrt(1 - 0.85^2)
= 0.52678
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sin φ

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[15] rho_T = rho_20 * (1 + alpha * (T_c - 20 degC) / (1 K))
= 0.01724 ohm·mm²/m × (1 + 0.00393 × (70 degC − 20 degC) / 1 K)
= 0.020628 ohm·mm²/m
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Resistivity at the operating temperature

[16] R = rho_T * L / (A_c * n_par)
= 0.020628 ohm·mm²/m × 80 m / (35 mm² × 1)
= 0.047149 ohm
Conductor resistance (one conductor, parallel paths combined)
[17] X = X_km * L / n_par
= 0.08 ohm/km × 80 m / 1
= 0.0064 ohm
Conductor reactance
[18] dV = sqrt(3) * I * (R * cos_phi + X * sin_phi)
= sqrt(3) × 100 A × (0.047149 ohm × 0.85 + 0.0064 ohm × 0.52678)
= 7.5254 V
Voltage drop
[19] dV_pct = 100 * dV / V_n
= 100 × 7.5254 V / 400 V
= 1.8814
Voltage drop, per cent
[20] P_loss = 3 * I^2 * R
= 3 × (100 A)^2 × 0.047149 ohm
= 1414.5 W
Conductor power loss
[21] u_dv = dV_pct / dv_limit
= 1.8814 / 5
= 0.37627
Drop utilisation against the limit
[25] u_Iz = I / I_z
= 100 A / 119 A
= 0.84034
Load against the current-carrying capacity
[26] T_c_est = T_amb + (T_rated - T_amb) * u_Iz^2 [1](App. 4 §6 form)
= 30 degC + (70 degC - 30 degC) × 0.84034^2
= 58.247 degC
Estimated conductor temperature

5. Results

SYMBOL	VALUE	DESCRIPTION
R	0.047149 ohm	Resistance
X	0.0064 ohm	Reactance
dV	7.5254 V	Voltage drop
dV_pct	1.8814	Voltage drop (%)
P_loss	1414.5 W	Power loss
T_c_est	58.247 degC	Estimated conductor temperature

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	I ≤ I_z	u_Iz = 0.84034	u_Iz ≤ 1	0.84	[1](523)
PASS	Voltage drop ≤ limit	u_dv = 0.37627	u_dv ≤ 1	0.376	[1](525)

7. Applicability

No warnings.

8. Verification

```
Template elec.voltage-drop@1.0.1 Content hash 2c07193ab8ffc91f Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

9. References

1. British Standards Institution; Institution of Engineering and Technology. Requirements for Electrical Installations: IET Wiring Regulations. BS 7671. IET.
2. Hughes E. Hughes Electrical and Electronic Technology. 12th ed. Pearson; 2016.